



taumonplot Documentation

Release 2.0.5

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July 01, 2014

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OVERVIEW

1.1 Description

taumonplot is a python module using matplotlib to plot TAU monitoring data.

1.2 Author information

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1.3 Disclaimer

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USAGE

2.1 Requirements

2.1.1 Programs:

- DLR TAU solver - to generate monitoring files

DLR TAU solver

The DLR TAU code is a fluid solver. See [description](http://tau.dlr.de/code-description/) (<http://tau.dlr.de/code-description/>) for more info.

2.1.2 Python packages:

- numpy >= version 1.6.1
- matplotlib >= version 0.99.1
- Tkinter >= revision 81008

Note:

Linux: All packages mentioned are available via the standard repositories and can be downloaded for example with a package manager

Windows: Recommended is the installation of a python distribution, that includes many other packages. An example is [PythonXY](https://code.google.com/p/pythonxy/wiki/Downloads) (<https://code.google.com/p/pythonxy/wiki/Downloads>).

To install the needed packages in your existing python distribution you can use the precompiled binaries (inofficial) on Christoph Gohlke's [page](http://www.lfd.uci.edu/~gohlke/pythonlibs/) (<http://www.lfd.uci.edu/~gohlke/pythonlibs/>) at the University of California, Irvine.

NumPy

[NumPy](https://pypi.python.org/pypi/numpy) (<https://pypi.python.org/pypi/numpy>) is an extension to the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large library of high-level mathematical functions to operate on these arrays.

Matplotlib

[Matplotlib](http://matplotlib.org/) (<http://matplotlib.org/>) is a plotting library for the Python programming language and its NumPy numerical mathematics extension.

Tkinter

Tkinter (<http://tcl.sourceforge.net/>) is Python's de-facto standard GUI (Graphical User Interface) package. It is a thin object-oriented layer on top of Tcl/Tk. Tkinter is included in python.

2.1.3 Getting ready

Enable TAU monitoring

Note: valid for TAU version 2013.2

To generate a monitoring file during a TAU run the user has to enable the monitoring parameter in the TAU input file.

```
Monitorhistory(0/1):integer,default0
```

- Turns on/off the print out of the Monitoring values in Tecplot ASCII format in an extra file. Whenever a restart is done and the Output files prefix is not changed the output of the values will be continued in the previous monitoring file. The monitoring files are named as prefix.monitoring.pval.dat and prefix.unsteady.monitoring.pval.dat in a steady and unsteady computation. If a restart is made with the turned on monitoring history the monitoring values have to be fixed over the whole computation.

Example Monitorhistory(0/1):1

The monitoring values are indicated with the following parameter:

```
Monitoringvalues:characterstring,default(none)
```

- Character string containing the names of the quantities to be monitored during execution of the solver. The names are separated by underline, e.g. Residual Max-res C-lift C-drag. The available names with a short explanation are listed by the solver in the standard output and are listed in Table 4 in the subsection Output files. Some of the monitoring values can be used to check the convergence of the inner iterations (parameter Use Cauchy convergence control). Those variables are marked with an "X" in the second column.

Example Monitoringvalues:Residual_C-lift_C-drag_C-my_Angle-a

See [Table of monitoring values](#) for available monitoring values.

2.1.4 Starting taumonplot

Please see the section [Manual](#) for information.

2.1.5 Table of monitoring values

Implemented monitoring values (to be separated by '_')	
KEY	COMMENT
Residual	norm of rho-increments
dvx/dt	norm of u-increments
dvy/dt	norm of v-increments
dvz/dt	norm of w-increments
dp/dt	norm of pressure-increments
drhoE/dt	norm of energy-increments
dnue/dt	norm of nue-increments
drk/dt	norm of rho_k-increments
Continued on next page	

Table 2.1 – continued from previous page

Implemented monitoring values (to be separated by ‘_’)	
KEY	COMMENT
drk2/dt	norm of rho_k2-increments
Max-res	maximum of rho-increments
X-max-res	grid x-coordinate of Max-res
Y-max-res	grid y-coordinate of Max-res
Z-max-res	grid z-coordinate of Max-res
Max-res-global	global maximum of rho-increments over all domains
C-lift	lift-coefficient
C-lift-p	pressure part of C-lift
C-lift-v	viscous part of C-lift
C-drag	drag-coefficient
C-drag-p	pressure part of C-drag
C-drag-v	viscous part of C-drag
C-sidef	coeff of sideforce
C-sidef-p	pressure part of C-sidef
C-sidef-v	viscous part of C-sidef
C-mx	rolling moment coefficient
C-mx-p	pressure part of rolling moment coefficient
C-mx-v	viscous part of rolling moment coefficient
C-my	pitching moment coefficient
C-my-p	pressure part of pitching moment coefficient
C-my-v	viscous part of pitching moment coefficient
C-mz	yawing moment coefficient
C-mz-p	pressure part of yawing moment coefficient
C-mz-v	viscous part of yawing moment coefficient
C-fx	force-coefficient (x-component)
C-fx-p	pressure part of force-coefficient (x-component)
C-fx-v	viscous part of force-coefficient (x-component)
C-fy	force-coefficient (y-component)
C-fy-p	pressure part of force-coefficient (y-component)
C-fy-v	viscous part of force-coefficient (y-component)
C-fz	force-coefficient (z-component)
C-fz-p	pressure part of force-coefficient (z-component)
C-fz-v	viscous part of force-coefficient (z-component)
Fx	x-component of force [N]
Fx-p	pressure part of x-component of force [N]
Fx-v	viscous part of x-component of force [N]
Fy	y-component of force [N]
Fy-p	pressure part of y-component of force [N]
Fy-v	viscous part of y-component of force [N]
Fz	z-component of force [N]
Fz-p	pressure part of z-component of force [N]
Fz-v	viscous part of z-component of force [N]
Mx	x-component of moment [Nm]
Mx-p	pressure part of x-component of moment [Nm]
Mx-v	viscous part of x-component of moment [Nm]
My	y-component of moment [Nm]
My-p	pressure part of y-component of moment [Nm]
My-v	viscous part of y-component of moment [Nm]
Mz	z-component of moment [Nm]
Mz-p	pressure part of z-component of moment [Nm]
Mz-v	viscous part of z-component of moment [Nm]
Min-mach	minimum Machnumber in field

Continued on next page

Table 2.1 – continued from previous page

Implemented monitoring values (to be separated by ‘_’)	
KEY	COMMENT
Max-mach	maximum Machnumber in field
Heatflow	total heatflow [W]
Res-heat	heatflux-increment
Min-y+	minimum value of y+
X-min-y+	grid x-coordinate of Min-y+
Y-min-y+	grid y-coordinate of Min-y+
Z-min-y+	grid z-coordinate of Min-y+
Max-y+	maximum value of y+
X-max-y+	grid x-coordinate of Max-y+
Y-max-y+	grid y-coordinate of Max-y+
Z-max-y+	grid z-coordinate of Max-y+
Min-kr	minimum value of kr
X-min-kr	grid x-coordinate of Min-kr
Y-min-kr	grid y-coordinate of Min-kr
Z-min-kr	grid z-coordinate of Min-kr
Max-kr	maximum value of kr
X-max-kr	grid x-coordinate of Max-kr
Y-max-kr	grid y-coordinate of Max-kr
Z-max-kr	grid z-coordinate of Max-kr
Min-kr+	minimum value of kr+
X-min-kr+	grid x-coordinate of Min-kr+
Y-min-kr+	grid y-coordinate of Min-kr+
Z-min-kr+	grid z-coordinate of Min-kr+
Max-kr+	maximum value of kr+
X-max-kr+	grid x-coordinate of Max-kr+
Y-max-kr+	grid y-coordinate of Max-kr+
Z-max-kr+	grid z-coordinate of Max-kr+
act-area	Area actuation bdry
act-targ-mf	Actuation target massflow
act-pres	Actuation bdry pressure
Min-eddyv	min dim-less eddy viscosity
Max-eddyv	max dim-less eddy viscosity
Min-k	min kinetic energy
Max-k	max kinetic energy
Min-k2	min 2nd turb. quantity
Max-k2	max 2nd turb. quantity
Max-mtm	max eddy-viscosity/laminar-viscosity
Angle-a	angle of attack alpha
Angle-b	yaw angle beta
Sideslip	sideslip angle beta
Farf-vx	farfield vx
Farf-vz	farfield vz
Mtm-2t	number of pseudo time steps (unsteady)
Alpha(t)	pitch amplitude
H(t)	heave amplitude [grid unit]
dalpha-dt	pitch angular velocity
dh-dt	heave velocity [grid unit/s]
S-k-e	structural kinetic energy
T/tperiod	time normalized with period length
Phi	rotation angle phi
Psi	rotation angle psi
Xi	rotation angle xi

Continued on next page

Table 2.1 – continued from previous page

Implemented monitoring values (to be separated by ‘_’)	
KEY	COMMENT
Trans-x	translation in x direction [grid unit]
Trans-y	translation in y direction [grid unit]
Trans-z	translation in z direction [grid unit]
Farf-mach	effective farfield Mach number at origin
Farf-alpha	effective farfield angle of attack alpha at origin [deg]
Farf-beta	effective farfield angle of attack beta at origin [deg]
les-dt	timescale resolution of des
les-percent	% of des points
hybcentpercent	% of fluxes computed with pure central
vort-center-x	center of vorticity (x)
vort-center-y	center of vorticity (y)
vort-center-z	center of vorticity (z)
vort-total	total vorticity
dudx	dudx
dvdy	dvdy
dwdz	dwdz
total-ke	total ke
mean-dissipl	molecular dissipation
rmsu	rms u
rmsv	rms v
rmsw	rms w
rmsp	rms pressure
skew	velocity derivative skewness
kurt	normalized velocity derivative kurtosis
Taylor-f	Longitudinal Taylor length scale
Taylor-g	Transverse Taylor length scale
Taylor-Re	Re based on Taylor scale
L2n-vc	L2 norm of vortical correction
Max-vc	Max value of vortical correction
Min-vc	Min value of vortical correction
Res-lift	cl-increment
Res-drag	cd-increment
R-time	Total CPU time (real)
U-time	Total CPU time (user)
Rhs-time	CPU time for one compute_residual() call.
Norm-time	Normalized CPU time (real)
dJ/da-1	Adjoint & Primal: Partial dJ/da
dJ/da-2	Adjoint: $\psi * \text{partial } dR/da$, Primal: $\text{Partial } dJ/dW * dW/da$
dJ/da	Adjoint & Primal: derivative of cost fn wrt design var.
Sens	Functional sensitivity
Freqdom-Cl-Amp	Freqdom: Amplitude of CL
Freqdom-Cl-Phase	Freqdom: Phase shift of CL
Freqdom-Cd-Amp	Freqdom: Amplitude of CD
Freqdom-Cd-Phase	Freqdom: Phase shift of CD
Freqdom-Cmx-Amp	Freqdom: Amplitude shift of CM_X
Freqdom-Cmx-Phase	Freqdom: Phase shift of CM_X
Freqdom-Cmy-Amp	Freqdom: Amplitude shift of CM_Y
Freqdom-Cmy-Phase	Freqdom: Phase shift of CM_Y
Freqdom-Cmz-Amp	Freqdom: Amplitude shift of CM_Z
Freqdom-Cmz-Phase	Freqdom: Phase shift of CM_Z
RPM-Res-P	RPM residual on Newton subspace
RPM-Res-Q	RPM residual on complementary subspace.
Continued on next page	

Table 2.1 – continued from previous page

Implemented monitoring values (to be separated by ‘_’)	
KEY	COMMENT
RPM-Dim-P	RPM Newton subspace dimension

MANUAL

3.1 Starting taumonplot

- run the python script directly in a Text editor (F5 in SciTE)
- run the python script from a terminal `pythontaumonplot_v2-04.py`
- run the function `create_plot(...)` of the module `taumonplot_v2-04` in another script
- **Monitoring data on a cluster:**
 1. Place the script on the cluster. Run the script with a Text editor on your own machine. You can use all options like auto updating. The monitoring file is transfered to your own machine for reading and display.or
 2. Place the script on the cluster. Run the script from you local terminal.

3.2 GUI

3.2.1 GUI overview

The GUI consists of two main parts. The first is a cavas that displays the figure with the monitoring data on top of the GUI. The second is a frame on the botom containing options for loading files, plotting etc.

3.2.2 Canvas

The canvas plots the monitoring data dependent on the choosen options.

3.2.3 Options

3.2.4 Tool tips

Hoovering over a widget (buttons, letter boxes etc.) will open a tool tip window next to the cursor, that shortly describes the functionality of the widget.

3.2.5 Programm options

Quit button	closes the program
Help button	opens license information and manual

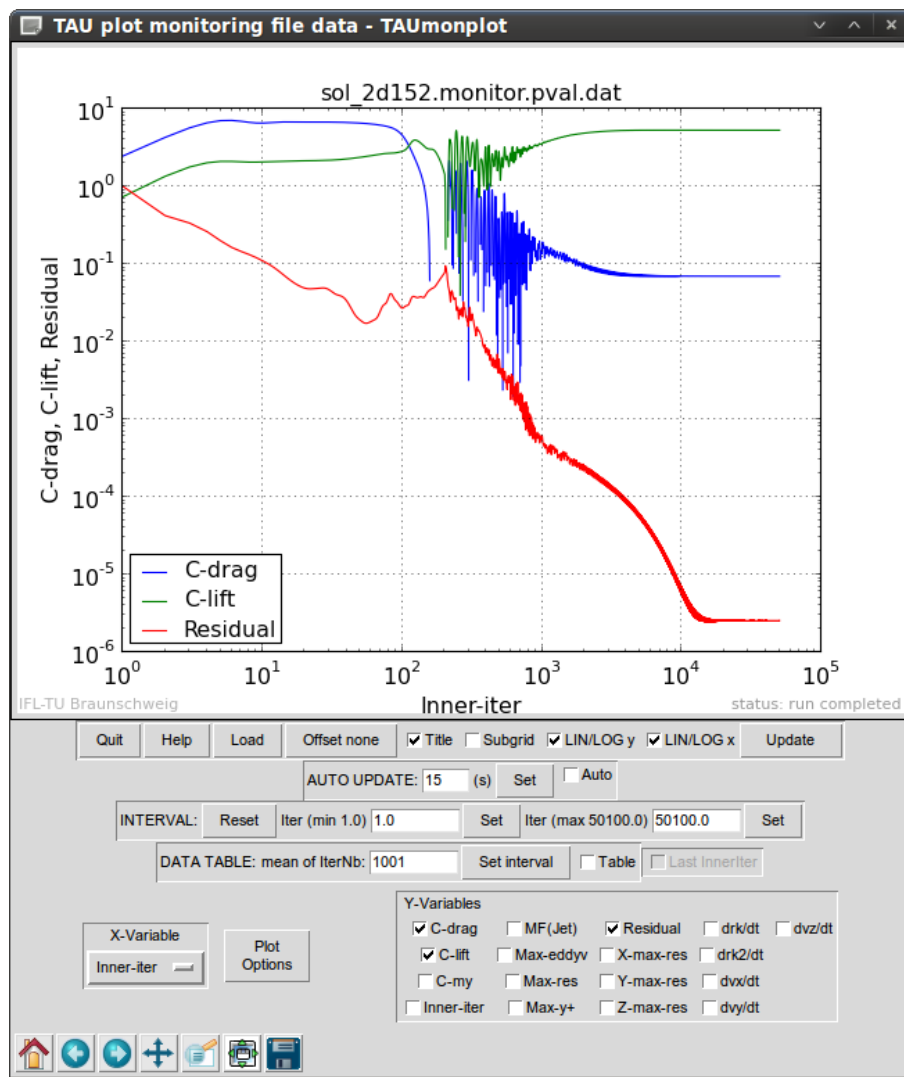


Figure 3.1: Complete taumonplot GUI. At the top is the plot canvas, at the bottom the options

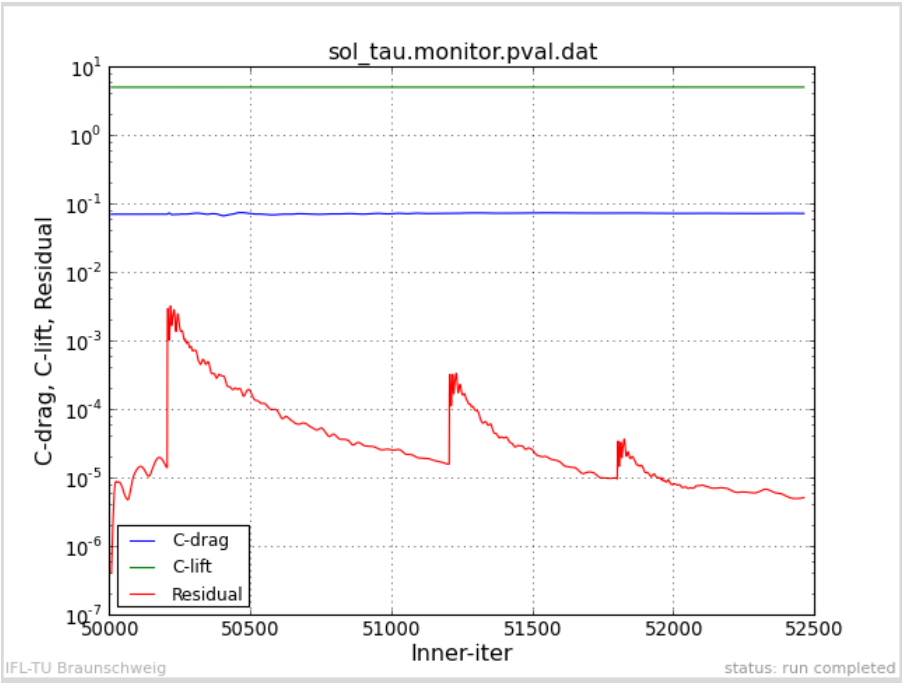


Figure 3.2: Matplotlib canvas

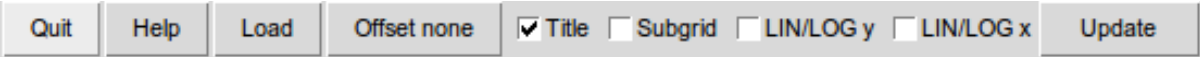


Figure 3.3: option bar

3.2.6 File options

Load button	opens a file dialog to open a different Tau monitoring file

3.2.7 Plot options

Title switch	toggles title display of monitoring file
Subgrid switch	toggles the display of subgrid lines for the logarithmic axes
LIN/LOGx switch	toggles the change between logarithmic or linear axis for the abscissa
LIN/LOGy switch	toggles the change between logarithmic or linear axis for the ordinate
Offset (none/on/math) button	toggles number offset of axis ticks (documentation (http://matplotlib.org/1.2.0/api/ticker_api.html))
Update button	updates the data from the monitoring file (new data is displayed if monitoring file has received an update from TAU)

3.2.8 Autoupdate option

AUTOUPDATE letter box	interval can be set here (in seconds)
Set button	sets the update interval from the letter box
Auto toggle	auto update on/off switch

Note: The script will execute an update command every *interval seconds* as if the user presses the Update button. If the file is too large and the interval is too small, this can lead to stability problems

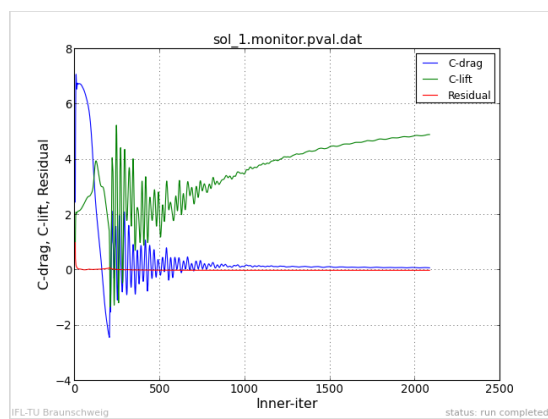


Figure 3.4: Canvas with linear X, linear Y axes

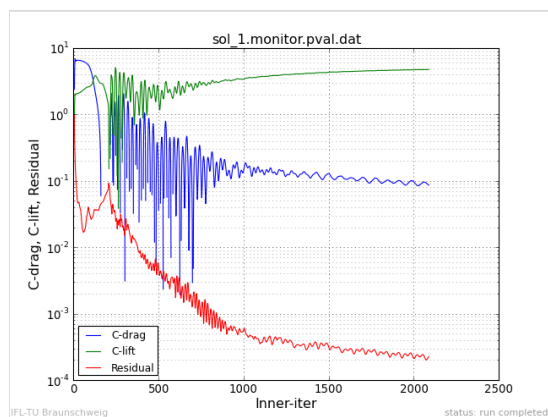


Figure 3.5: Canvas with linear X, logarithmic Y axes - with subgrid

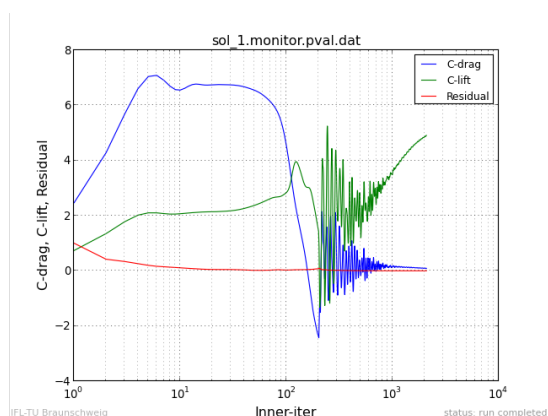


Figure 3.6: Canvas with logarithmic X, linear Y axes - with subgrid

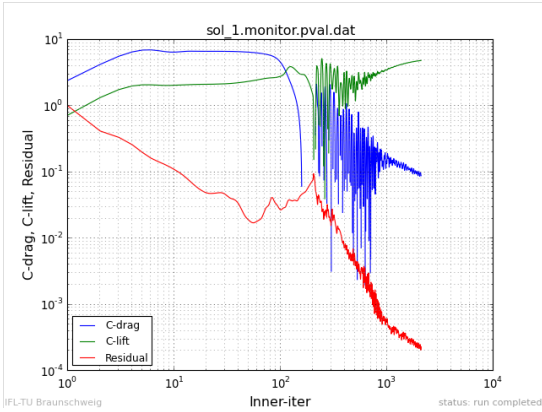


Figure 3.7: Canvas with logarithmic X, logarithmic Y axes - with subgrid

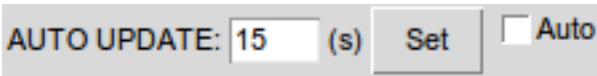


Figure 3.8: autoupdate option bar

3.2.9 Iteration interval display options



Figure 3.9: plot interval option bar

Reset button	resets the minimum and maximum values to display to the complete range
Iter/Time (minXXXXXX) letter box	number of start iteration/time to plot
Set button	sets the start iteration/time to plot
Iter/Time (maxXXXXXX) letter box	number of end iteration/time to plot
Set button	sets the end iteration/time to plot

3.2.10 Data table display options

DATATABLE : meanofIterNb letter box	number of iterations to compute mean for data table
Table toggle	toggle the display of a data table on the plot with mean values of the last iterations
Setinterval button	sets the range of last iterations to include in the mean value (standard values is 1/50th of iteration range at startup)

3.2.11 Time: Plot last inner iteration value

When plotting transient data over the time variable, the value of the inner iterations is often not of interest. To plot only the last values of the inner iterations, which are the result of the time step, a switch has been implemented. The switch is only active when X-Variable is `thistime`.



Figure 3.10: data table option bar

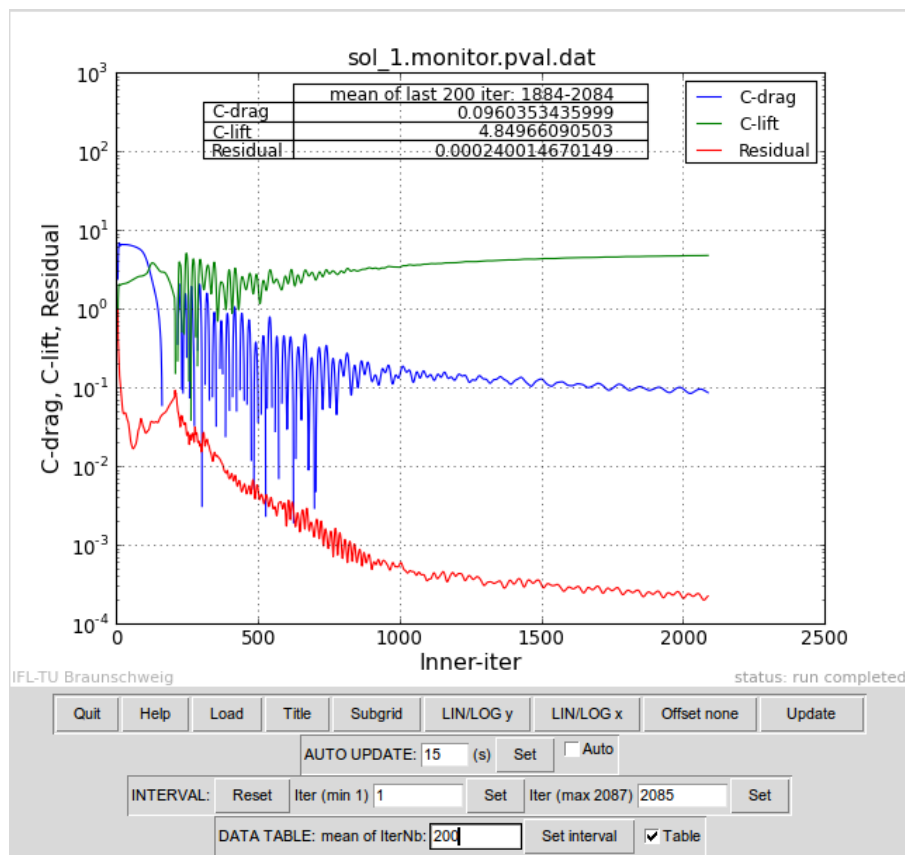


Figure 3.11: Overlaid data table, mean computation set to 200 iterations

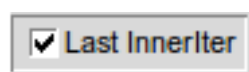


Figure 3.12: Last InnerIter switch

LastInnerIter switch	toggles to plot all iterations or only the last value of the inner iterations (switch active with X-Variable=thistime)

Example of effect:

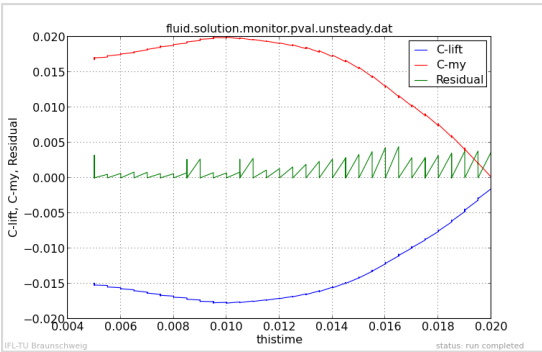


Figure 3.13: Last inner Iter off

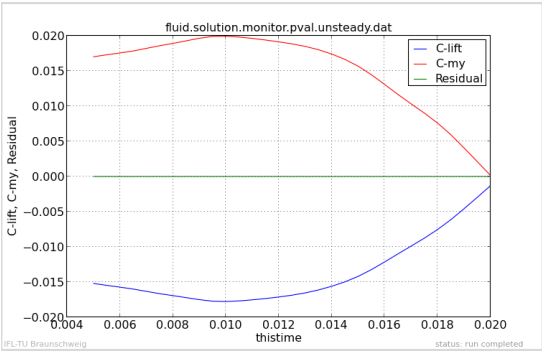


Figure 3.14: Last inner Iter on

3.2.12 Variable choice options

The variable choices will be automatically adapted, depending on the variables in monitoring file.

X-Variable drop-down menu	one choice of variable for the abscissa.

Choosing a variable different from Inner-iter will deactivate the [Iteration interval display options](#) and [Data table display options](#). Another sensible variable to choose for the abscissa is the *this-time* variable for transient monitoring files.

Y-variables checkboxes	multiple choices of variables for the ordinate

3.2.13 Plot options

This button opens another windows which contains the plot options for the variables and axes. The tabl is generated at startup for the loaded variables.

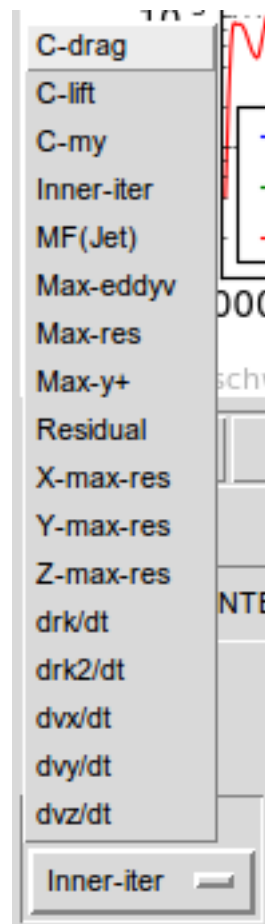


Figure 3.15: x-variable drop down menu

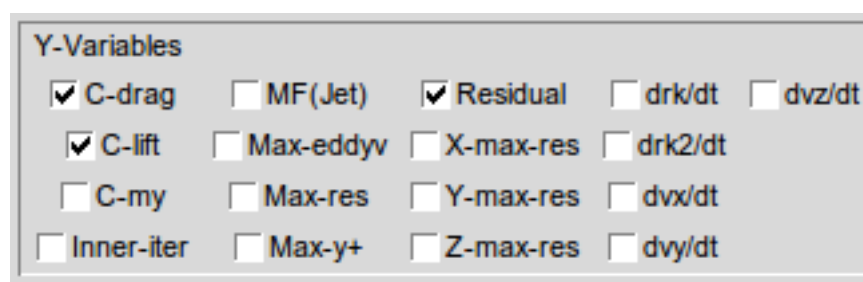


Figure 3.16: y-variable checkboxes

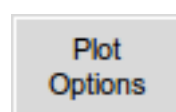


Figure 3.17: Plot options button

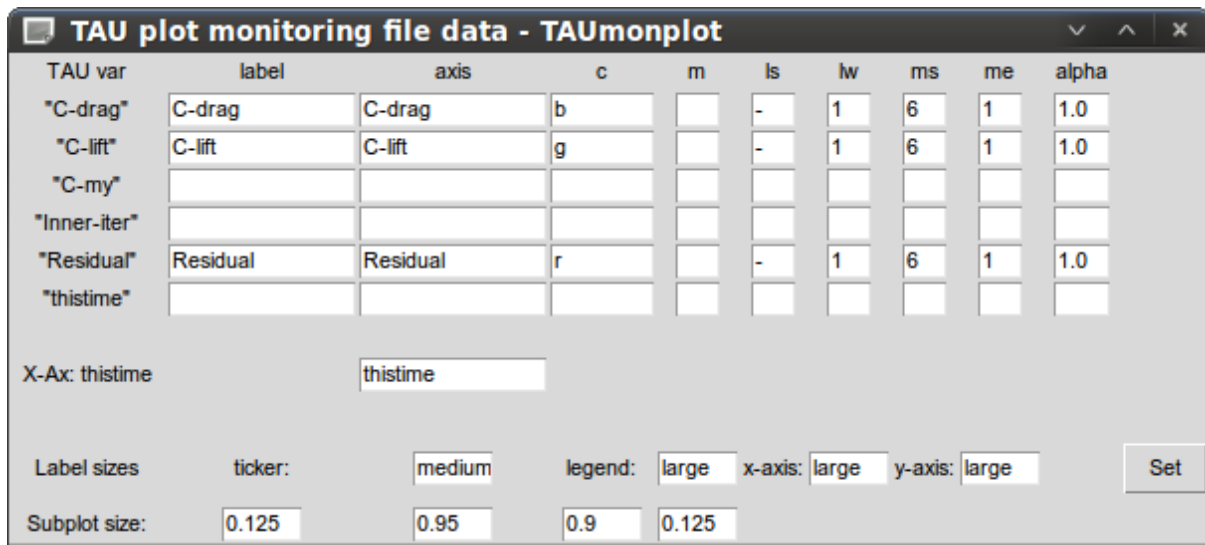


Figure 3.18: Plot options

Column	Option
TAU var	name of the loaded TAU variable (see Table of monitoring values)
label	label name as it appears in the legend. Latex formatting possible with \$\$
axis	label name as it appears on the y-axis. Latex formatting possible with \$\$
c	Color of the curve: standard colors, grey shades, html hex strings and html names for colors possible
m	Marker ('o','h','^','v','<','>','_','1','2','3','4','8','p','l','d',';','+','s','*','l','x','D','H','.'))
ls	Line style ('-', '--', '...', '-.', '...', 'o','^','v','<','>','s','+','x','d','1','2','3','4','h','p','l','S','H')
lw	Line width (float)
ms	Marker size (float)
me	Marker every n-th value
alpha	Alpha blending value (0.0 transparent through 1.0 opaque)

X-Axis: <i>variable</i>	label name as it appears on the x-axis. Latex formatting possible with \$\$
-------------------------	---

Subplot size	Subplot distance from the respective borders - see hover tooltip
--------------	--

3.2.14 Matplotlib figure options



Figure 3.19: Matplotlib figure option bar

- (home button? : no functionality)
- (back button? : no functionality)
- (forward button? : no functionality)
- Pan/Zoom button : allows panning and zooming of the image
- Zoomrect button : allows zoom into a rectangle
- Options button : to adjust subplot params. See documentation for [subplots_adjust](http://matplotlib.org/api/figure_api.html#matplotlib.figure.Figure.subplots_adjust) (http://matplotlib.org/api/figure_api.html#matplotlib.figure.Figure.subplots_adjust).
- Savefigure button : allows saving of the canvas

3.3 File choice

3.3.1 Autochoice

1. Execute taumonplot
2. The program will search in the current working directory for a file with the endings
 - `‘.monitor.tmp.dat’`
 - `‘.monitor.pval.dat’`
 - `‘.monitor.tmp.unsteady.dat’`
 - `‘.monitor.pval.unsteady.dat’`

and the `name_prefix=‘fluid.solution’`.

If a file is found, it is opened.

3.3.2 Choice by file dialog

1. Same behaviour as in [Autochoice](#)
2. If no file is found, a file dialog is opened, that allows the user to choose any file for input

3.3.3 Choice by filename and subdirectory

1. Open the taumonplot script file
2. Declare `subdirectory` and `name_prefix` of tau monitoring file (without the endings `‘.monitor.tmp.dat’`, `‘.monitor.pval.dat’`, `‘.monitor.tmp.unsteady.dat’`, `‘.monitor.pval.unsteady.dat’`)
3. Verify that `data_line` and `title_line` are correct
4. Run program

3.3.4 Choice by function call

```
from taumonplot import create_plot
#
subdirectory = './'
name_prefix = 'fluid.solution' # name of solution file
data_line = 25    # line where data values start in monitoring file
title_line = 23   # line with variable names
# start variables to plot
x_variable_list = ['Inner-iter']
y_variable_list = ['Residual',
                  'C-lift',
                  'C-drag' ]
check_iter_numbers = True    # Enable to check ascending order of iteration numbers
#
create_plot(subdirectory,
            name_prefix,
            data_line,
            title_line,
            x_variable_list,
            y_variable_list,
            check_iter_numbers)
#
```

Function `create_plot(subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list, check_iter_numbers, save_to_file=False)`

Parameters

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [list of string] for x (display automatically at startup), Example:
`x_variable_list=[{}'Inner-iter']`

y_variable_list [list of strings] for y (display automatically at startup), Example:
`y_variable_list=[{}'Residual','C-lift','C-drag' ]`

check_iter_numbers [boole] Check ascending order of iteration numbers

save_to_file [bool] save to file (deprecated)

3.4 Information for using taumonplot

- the user has to press the `Update` button to reload the monitoring file
- all changes to the plot via other buttons do **not** trigger a reload of the monitoring file
- the status of the TAU run is indicated in the buttom righth corner. This status does not indicate the status of the TAU process itself but gets its information from the file ending, which is read. The running TAU solver creates a file with the endings `'monitor.tmp.dat'` (steady state run) or `'monitor.tmp.unsteady.dat'` (dual time stepping - unsteady run). While a file with this file ending is loaded, *taumonplot* will indicate the run as not completed. TAU will change the filename on its own after it completed to `'monitor.pval.dat'` or `'monitor.pval.unsteady.dat'`. However, when the solver fails to complte its run, the file endings are not changed. An indication for a failed run is no change in monitoring data after several seconds (normal computations) or minutes (very large computations).

VERSION INFORMATION

4.1 Version table

version	date	changes
2.05	17.02.2014	subplot border distance adjustment in plot options
2.04	17.02.2014	major: added plot options, direct manipulation of plot lines, axis, legend
2.03	14.02.2014	changed title, subgrid, lin/log buttons to checkbuttons
2.02	13.02.2014	added interval control for thistime
2.01	13.02.2014	added switch to plot only last value of an inner iteration, when plotting x-value 'thistime'
2.00	22.03.2013	adapted model name for Sphinx from 'taumonplot_vX-YZ.py' to 'taumonplot_vX_YZ.py'
2.00	22.03.2013	added Sphinx compatible documentation to source code
2.00	22.03.2013	created Sphinx documentation: html, pdf
1.98	22.03.2013	keeps lower interval during updates, only adjusts the upper interval
1.97	16.03.2013	load file path set identical to initial subdirectory
1.96	10.01.2013	modified by Ian Krukow
1.96	10.01.2013	numbers of inner iterations modified
1.96	10.01.2013	for the case that the iterations are counted separately for each timestep
1.95	15.12.2012	debug load error
1.95	15.12.2012	auto update: Tau run finished: load appropriate file and end auto update
1.94	07.12.2012	circumvent error with no y-variables selected
1.93	02.12.2012	auto update switch and time
1.93	02.12.2012	remodeled GUI buttons
1.93	02.12.2012	added Tooltips
1.92	17.10.2012	pop-up when file not available
1.91	10.10.2012	fixed Update button not requiring two clicks for update
1.91	10.10.2012	fixed program exit on clicking 'cancel' in the load file dialogue
1.90	28.09.2012	renamed plot_monitoring to taumonplot
1.90	28.09.2012	integrated code into Plotcreator, speed up reload data
1.90	28.09.2012	terminal startup support with filename
1.85	27.08.2012	adjustment to data table: displays mean range, maximum of mean range is
1.85	27.08.2012	now tied to maximum value to display, thus allowing mean of desired ranges
1.80	24.08.2012	x-Variable drop down menu
1.75	12.06.2012	tick formater for offset
1.70	21.05.2012	bugfix data range entries
1.65	15.05.2012	bugfix table data with no values
1.65	15.05.2012	fix update of table range values
1.60	06.05.2012	added dialog to ask for file if file not available
1.60	06.05.2012	added Load button to load other files
1.55	06.05.2012	added choice to display a data table of all selected y-variables
1.55	06.05.2012	values represent the mean of selected last number of iterations
1.55	06.05.2012	added manual in header

Continued on next page

Table 4.1 – continued from previous page

version	date	changes
1.55	06.05.2012	added popup before quit
1.50	04.05.2012	legend location changed to best
1.50	04.05.2012	added choice of iteration range to display
1.45	14.04.2012	toggle major, minor grid lines
1.45	14.04.2012	bugfix save .png
1.40	12.04.2012	added major grid lines
1.35	07.04.2012	fixed reset of start values when updating or toggling buttons
1.35	07.04.2012	added title_line variable
1.30	20.03.2012	added unsteady monitoring
1.30	20.03.2012	removed dependency on read_ascii_module
1.25	16.03.2012	toggle logarithmic scales
1.20	16.03.2012	cleaned GUI
1.15	15.03.2012	interactive choice of x and y-values
1.10	15.03.2012	integration in Tkinter, interactive reload
1.05	15.03.2012	monitor file name change when run completed
1.00	08.03.2012	first implementation

TODO

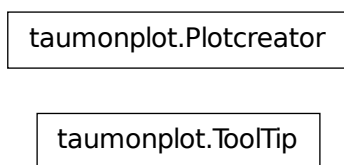
This section lists ideas for adaptations/features.

- **add additional y-axis, Y-variable-choice for axis, additional lin/log-toggle**
 - buttons to remove status and branding
 - reset button for plot options
- adapt source code to full object oriented approach
- when loading another file remap the X- and Y-Variable entries

TAUMONPLOT

6.1 Module taumonplot

6.1.1 Inheritance diagram



6.1.2 Summary

Classes and members

<code>taumonplot.Plotcreator(subplot, ...[, ...])</code>	Class to create a matplotlib figure plot
<code>taumonplot.Plotcreator.__init__ _(subplot, ...)</code>	Create a matplotlib figure plot
<code>taumonplot.Plotcreator.__module__</code>	<code>str(object) -> string</code>
<code>taumonplot.Plotcreator.add_TAU_ status(fig)</code>	Adds the current TAU solver status to the canvas.
<code>taumonplot.Plotcreator.get_log_ name(...)</code>	Returns the logfile name or error when file not found
<code>taumonplot.Plotcreator.get_ variables()</code>	Returns a list of available variables and the min and max value of the x variable from the TAU monitoring file
<code>taumonplot.Plotcreator.plot_ monitor_file(subplot)</code>	Uses read data and adds it to the subplot.
<code>taumonplot.Plotcreator. replot(subplot, ...)</code>	Executes plot_monitor_file method

taumonplot.Plotcreator

```
class taumonplot.Plotcreator (subplot, subdirectory, name_prefix, data_line, title_line,
                             x_variable_list, y_variable_list, check_iter_numbers,
                             save_to_file=False)
```

Class to create a matplotlib figure plot

```
__init__ (subplot, subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list,
          check_iter_numbers, save_to_file=False)
Create a matplotlib figure plot
```

Parameters

subplot [matplotlib subplot instance] the figure subplot

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

check_iter_numbers [bool] check ascending order of iteration numbers

save_to_file [bool] switch to save file (deprecated)

Methods

<code>__init__(subplot, subdirectory, name_prefix, ...)</code>	Create a matplotlib figure plot
<code>add_TAU_status(fig)</code>	Adds the current TAU solver status to the canvas.
<code>get_log_name(subdirectory, name_prefix)</code>	Returns the logfile name or error when file not found
<code>get_variables()</code>	Returns a list of available variables and the min and max value of the x variable from the TAU monitoring file
<code>plot_monitor_file(subplot[, data_line, ...])</code>	Uses read data and adds it to the subplot.
<code>replot(subplot, data_line, title_line, ...)</code>	Executes plot_monitor_file method

taumonplot.Plotcreator.__init__

```
Plotcreator.__init__ (subplot, subdirectory, name_prefix, data_line, title_line, x_variable_list,
                      y_variable_list, check_iter_numbers, save_to_file=False)
```

Create a matplotlib figure plot

Parameters

subplot [matplotlib subplot instance] the figure subplot

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

check_iter_numbers [bool] check ascending order of iteration numbers

save_to_file [bool] switch to save file (deprecated)

taumonplot.Plotcreator.__module__

`Plotcreator.__module__ = 'taumonplot'`

taumonplot.Plotcreator.add_TAU_status

`Plotcreator.add_TAU_status (fig)`

Adds the current TAU solver status to the canvas.

Parameters

fig - matplotlib figure instance the figure

taumonplot.Plotcreator.get_log_name

`Plotcreator.get_log_name (subdirectory, name_prefix)`

Returns the logfile name or error when file not found

Parameters

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

taumonplot.Plotcreator.get_variables

`Plotcreator.get_variables ()`

Returns a list of available variables and the min and max value of the x variable from the TAU monitoring file

Parameters none

taumonplot.Plotcreator.plot_monitor_file

`Plotcreator.plot_monitor_file (subplot, data_line=1, title_line=23, x_variable_list=['Inner-iter'], y_variable_list=['Residual'], save_to_file=False)`

Uses read data and adds it to the subplot.

It additionally creates image files when save_to_file is set to True (obsolete) .

Parameters

subplot [matplotlib subplot instance] the figure subplot

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

save_to_file [bool] switch to save file (deprecated)

taumonplot.Plotcreator.replot

`Plotcreator.replot` (*subplot*, *data_line*, *title_line*, *x_variable_list*, *y_variable_list*, *save_to_file*)
Executes `plot_monitor_file` method

Parameters

- subplot** [matplotlib subplot instance] the figure subplot
- data_line** [integer] line number where data starts
- title_line** [integer] line number of variable names
- x_variable_list** [] list of strings for x-variable
- y_variable_list** [list of strings] list of strings for y-variable
- save_to_file** [bool] switch to save file (deprecated)

Module functions

<code>_is_string_like(obj)</code>	Check whether obj behaves like a string.
<code>ascii_read(filename, data_line[, ...])</code>	A wrapper around numpy loadtxt to load acsii data files, allowing for headers, string columns and returning the output as a dictionary using a line in the header as keys.
<code>createToolTip(widget, text)</code>	Function to create ToolTips given a widget and a Tooltip-Text
<code>create_plot(subdirectory, name_prefix, ...)</code>	Creates a Tkinter GUI for TAU monitoring file data

taumonplot._is_string_like

`taumonplot._is_string_like` (*obj*)
Check whether obj behaves like a string.

taumonplot.ascii_read

`taumonplot.ascii_read` (*filename*, *data_line*, *title_line=None*, *title_line_junk='#'*, *column_titles=True*, *return_header=False*, *string_cols=[]*, *usecols=[]*, *loadtxt_args=None*, *check_iter_numbers=False*)
A wrapper around numpy loadtxt to load acsii data files, allowing for headers, string columns and returning the output as a dictionary using a line in the header as keys.

Parameters

- filename** [str] full filename of file e.g. “swicsdata.txt”, or file object
- data_line** [int] line number for the first line of data
- title_line** [int, optional] line in the file where the list of coloum titles are (default 2 above `data_line`)
- title_line_junk** [str, optional] A string at the begging of the title line which should be removed also passed to loadtxt as comments kwarg
- column_titles** [bool, optional] If false will not return dict it returns ndarray.
- return_header** [bool, optional] Returns header lines as list if True
- string_cols** [list, optional] A sequence of column numbers eg, [1,2] which are strings, and will be extracted from the file sepatarly from the main loadtxt call.

usecols [list, optional] same as loadtxt_args{"usecols",[]}, selects specific columns from the data file. If using string_cols as well, usecols should include the columns that are in string_cols.

loadtxt_args: dict, optional Keyword Arguments to pass to numpy loadtxt

Created on Thu Feb 10 15:19:13 2011

@author: Stuart Mumford

This ascii_read a wrapper for loadtxt

DEV:

TO-DO

- Add mutiple Title pulls, for many columns
- Allow only string cols
- Improve logical flow of program (reduce if statements, esp if string_cols:)
- Include support for open file object

taumonplot.createToolTip

`taumonplot.createToolTip (widget, text)`
Function to create ToolTips given a widget and a TooltipText

taumonplot.create_plot

`taumonplot.create_plot (subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list, check_iter_numbers, save_to_file=False)`
Creates a Tkinter GUI for TAU monitoring file data

Parameters

subdirectory [string] directory containing monitoring data, absolute or relative
name_prefix [string] name prefix of monitoring file
data_line [integer] line number where data starts
title_line [integer] line number of variable names
x_variable_list [] list of strings for x-variable
y_variable_list [list of strings] list of strings for y-variable
check_iter_numbers [bool] check ascending order of iteration numbers
save_to_file [bool] switch to save file (deprecated)

6.1.3 Detailed Description

class `taumonplot.Plotcreator (subplot, subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list, check_iter_numbers, save_to_file=False)`

Class to create a matplotlib figure plot

__init__ (`subplot, subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list, check_iter_numbers, save_to_file=False`)
Create a matplotlib figure plot

Parameters

subplot [matplotlib subplot instance] the figure subplot

subdirectory [string] directory containing monitoring data, absolute or relative
name_prefix [string] name prefix of monitoring file
data_line [integer] line number where data starts
title_line [integer] line number of variable names
x_variable_list [] list of strings for x-variable
y_variable_list [list of strings] list of strings for y-variable
check_iter_numbers [bool] check ascending order of iteration numbers
save_to_file [bool] switch to save file (deprecated)

__module__ = 'taumonplot'

add_TAU_status (*fig*)

Adds the current TAU solver status to the canvas.

Parameters

fig - matplotlib figure instance the figure

get_log_name (*subdirectory, name_prefix*)

Returns the logfile name or error when file not found

Parameters

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

get_variables ()

Returns a list of available variables and the min and max value of the x variable from the TAU monitoring file

Parameters none

plot_monitor_file (*subplot, data_line=1, title_line=23, x_variable_list=['Inner-iter'], y_variable_list=['Residual'], save_to_file=False*)

Uses read data and adds it to the subplot.

It additionally creates image files when save_to_file is set to True (obsolete) .

Parameters

subplot [matplotlib subplot instance] the figure subplot

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

save_to_file [bool] switch to save file (deprecated)

replot (*subplot, data_line, title_line, x_variable_list, y_variable_list, save_to_file*)

Executes plot_monitor_file method

Parameters

subplot [matplotlib subplot instance] the figure subplot

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

save_to_file [bool] switch to save file (deprecated)

class taumonplot.**ToolTip** (*widget*)

Bases: object

ToolTip classs from Michael Foord http://www.voidspace.org.uk/python/weblog/arch_d7_2006_07_01.shtml#e387

__dict__ = dict_proxy({'__module__': 'taumonplot', 'hidetip': <function hidetip at 0x3828d70>, 'showtip': <functi

__init__ (*widget*)

__module__ = 'taumonplot'

__weakref__

list of weak references to the object (if defined)

hidetip ()

Remove text in tooltip window

showtip (*text*)

Display text in tooltip window

taumonplot.ascii_read (*filename*, *data_line*, *title_line=None*, *title_line_junk='#'*, *column_titles=True*, *return_header=False*, *string_cols=[]*, *usecols=[]*, *loadtxt_args=None*, *check_iter_numbers=False*)

A wrapper around numpy loadtxt to load acsii data files, allowing for headers, string columns and returning the output as a dictionary using a line in the header as keys.

Parenters

filename [str] full filename of file e.g. "swicsdata.txt", or file object

data_line [int] line number for the first line of data

title_line [int, optional] line in the file where the list of coloum titles are (default 2 above data_line)

title_line_junk [str, optional] A string at the begging of the title line which should be removed also passed to loadtxt as comments kwarg

column_titles [bool, optional] If false will not return dict it returns ndarray.

return_header [bool, optional] Returns header lines as list if True

string_cols [list, optional] A sequence of column numbers eg, [1,2] which are strings, and will be extracted from the file seperatly from the main loadtxt call.

usecols [list, optional] same as loadtxt_args{"usecols",[]}, selects specific columns from the data file. If using string_cols as well, usecols should include the columns that are in string_cols.

loadtxt_args: dict, optional Keyword Arguments to pass to numpy loadtxt

Created on Thu Feb 10 15:19:13 2011

@author: Stuart Mumford

This ascii_read a wrapper for loadtxt

DEV:

TO-DO

- Add mutiple Title pulls, for many columns
- Allow only string cols
- Improve logical flow of program (reduce if statements, esp if string_cols:)
- Include support for open file object

`taumonplot.createToolTip(widget, text)`

Function to create ToolTips given a widget and a TooltipText

`taumonplot.create_plot(subdirectory, name_prefix, data_line, title_line, x_variable_list, y_variable_list, check_iter_numbers, save_to_file=False)`

Creates a Tkinter GUI for TAU monitoring file data

Parameters

subdirectory [string] directory containing monitoring data, absolute or relative

name_prefix [string] name prefix of monitoring file

data_line [integer] line number where data starts

title_line [integer] line number of variable names

x_variable_list [] list of strings for x-variable

y_variable_list [list of strings] list of strings for y-variable

check_iter_numbers [bool] check ascending order of iteration numbers

save_to_file [bool] switch to save file (deprecated)

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Institute of Aircraft Design & Lightweight Structures (IFL), Technische Universitaet Braunschweig

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